Direct comparison test — 5.4

Direct Comparison Test: If $0 \le a_n \le b_n$, then

if $\sum_{n=1}^{\infty} b_n$ converges, so does $\sum_{n=1}^{\infty} a_n$

if $\sum_{n=1}^{\infty} a_n$ diverges, so does $\sum_{n=1}^{\infty} b_n$.

Example: Does $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ diverge or converge?

Observe that $0 \le \frac{1}{n^2+1} \le \frac{1}{n^2}$, so

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (by the p-test) implies that

$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges by the DCT.

Example: Does $\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$ diverge or converge?

$0 \le \ln(n) \le n$, so $0 \le \frac{1}{n} \le \frac{1}{\ln(n)}$. Then $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges (p-test), so also $\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$ diverges by the DCT.

Example: Does $\sum_{n=1}^{\infty} \frac{3+\sin(10n^5+2)}{n^3}$ converge or diverge?

$$\frac{2}{n^3} \le \frac{3+\sin(10n^5+2)}{n^3} \le \frac{4}{n^3}$$ because $|\sin(x)| \le 1$,

so $\sum_{n=1}^{\infty} \frac{4}{n^3}$ converging (p-test) implies that $\sum_{n=1}^{\infty} \frac{3+\sin(10n^5+2)}{n^3}$ converges by the DCT.

Example: Does $\sum_{n=1}^{\infty} \frac{n+1}{n^2}$ converge or diverge?

$\frac{n+1}{n^2} \geq \frac{1}{n}$ means that $\sum_{n=1}^{\infty} \frac{n+1}{n^2}$ diverges by the DCT since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges by the p-test.

What if we looked at $\sum_{n=1}^{\infty} \frac{n-1}{n^2}$ instead?

$\frac{n-1}{n^2} \not\geq \frac{1}{n}$

$\frac{n-1}{n^2} \leq \frac{1}{n}$ "smaller than something divergent"

<u>But</u> the <u>vibe</u> is still similar to $\frac{1}{n}$, so probably diverges.